DATABASE

MARKING SCHEME

Question 1 (2 marks)

Answer	Marks
Normalization is the process of organising data in a database into well-structured tables by applying a set of rules (normal forms).	1
Main purpose: to reduce data redundancy (duplication) and eliminate anomalies, thereby improving data integrity and consistency.	1
Total	**2**

Question 2 (2 marks)

Answer	Marks
Insertion anomaly: a new record cannot be added without adding unrelated data (e.g. a new product cannot be stored until an order exists for it).	
Update anomaly: the same data is stored in many rows, so changing it requires updating every row, otherwise the data becomes inconsistent.	

Deletion anomaly: deleting a record unintentionally removes other useful data stored in the same row.	
All three correctly explained = 2 marks; two correctly explained = 1 mark; one correctly explained = ½ mark	2
Total	**2**

Question 3 (2 marks)

Answer	Marks
CHAR stores fixed-length text (shorter values are padded with spaces), while VARCHAR stores variable-length text using only the space needed.	1
For prices, use DECIMAL (or NUMERIC), e.g. DECIMAL(10,2), because it stores exact numeric values with decimal places and supports arithmetic calculations. (Accept FLOAT/DOUBLE with a valid reason; do not accept CHAR or VARCHAR.)	1
Total	**2**

Question 4 (2 marks)

Award ½ mark for each correct point (two advantages and two disadvantages).

Advantages (any 2)	Marks
Speeds up data retrieval and searching (SELECT queries)	½
Speeds up sorting and joining of tables	½
Improves performance of queries with WHERE and ORDER BY	
Can enforce uniqueness (unique indexes)	

Disadvantages (any 2)	Marks
Takes up extra storage space	½
Slows down INSERT, UPDATE and DELETE operations because the index must be updated	½
Requires maintenance and planning	
Total	**2**

Question 5 (2 marks)

Answer	Marks
PRIMARY KEY: uniquely identifies each row in a table; only one per table; cannot contain NULL values.	1
UNIQUE constraint: ensures all values in a column are different; a table can have several; allows NULL values (depending on the DBMS).	1

Total	**2**

Question 6 (4 marks)

a) (1 mark)

Answer	Marks
The Sports column contains more than one value in a single cell (it is not atomic / has repeating groups). 1NF requires each cell to hold a single value.	1
Subtotal	**1**

b) (3 marks)

Answer	Marks
Table converted so that each cell holds one value (rows repeated for each sport)	1
All six rows correct with no data lost or added	1
Correct primary key: composite key {MemberID, Sport}	1
Subtotal	**3**

Model answer:

CLUB_MEMBER (MemberID, MemberName, Sport)

MemberID	MemberName	Sport
1	Mary	Football
1	Mary	Netball
2	John	Chess
3	Peter	Football
3	Peter	Chess
3	Peter	Swimming

Primary key: {MemberID, Sport}

(Accept an alternative that creates a separate MEMBER_SPORT table with key {MemberID, Sport}, provided the data is complete and correct.)

Question 6 Total	4

Question 7 (5 marks)

a) (2 marks)

Answer	Marks

Functional dependencies: {OrderID, ProductID} → Quantity; ProductID → ProductName, UnitPrice	1
Partial dependency: ProductName and UnitPrice depend on only part of the composite key (ProductID) rather than on the whole key	1
Subtotal	**2**

b) (2 marks)

Answer	Marks
ORDER_ITEM (OrderID, ProductID, Quantity), primary key {OrderID, ProductID}; ProductID is a foreign key	1
PRODUCT (ProductID, ProductName, UnitPrice), primary key ProductID	1
Subtotal	**2**

Model answer:

PRODUCT

ProductID	ProductName	UnitPrice
P1	Laptop	60000
P2	Mouse	800

ORDER_ITEM

OrderID	ProductID	Quantity
O1	P1	2
O1	P2	5
O2	P1	13

c) (1 mark)

Award 1 mark for any one anomaly correctly explained:

- **Update:** the price of the Laptop is stored in several rows, so a price change must be made in every row or the data becomes inconsistent.
- **Insertion:** a new product cannot be recorded until it is part of an order.
- **Deletion:** deleting the row for O1/P2 removes the only record of the Mouse and its price.

Question 7 Total 5

Question 8 (5 marks)

a) CREATE TABLE (3 marks)

Answer	Marks
Correct table name and columns with correct data types (StudentID INT, CourseID VARCHAR(6) or CHAR(6), EnrollDate DATE, Marks INT) and DEFAULT set to the current date	1
Correct CHECK constraint on Marks (CHECK (Marks BETWEEN 0 AND 100) or Marks >= 0 AND Marks <= 100)	1
Composite primary key PRIMARY KEY (StudentID, CourseID) and both foreign keys referencing STUDENT(StudentID) and COURSE(CourseID)	1
Subtotal	**3**

Model answer:

```
CREATE TABLE ENROLLMENT (
   StudentID  INT,
   CourseID   VARCHAR(6),
   EnrollDate DATE DEFAULT CURRENT_DATE,
   Marks      INT CHECK (Marks BETWEEN 0 AND 100),
   PRIMARY KEY (StudentID, CourseID),
   FOREIGN KEY (StudentID) REFERENCES STUDENT(StudentID),
   FOREIGN KEY (CourseID)  REFERENCES COURSE(CourseID)
```

);

(Accept DEFAULT (CURRENT_DATE) in MySQL, DEFAULT GETDATE() in SQL Server and DEFAULT SYSDATE in Oracle.)

b) CREATE INDEX (2 marks)

Answer	Marks
CREATE INDEX idx_student_name ON STUDENT(StudentName); (any valid index name)	1
Reason: the column is frequently used for searching, filtering (WHERE), sorting (ORDER BY) or joining, so an index speeds up these queries	1
Subtotal	**2**

Question 8 Total 5

Question 9 (6 marks)

a) (2 marks)

Answer	Marks

SELECT StudentName, Department FROM STUDENT WHERE Department = 'ICT' (correct columns, table and condition)	1
ORDER BY StudentName (ascending/alphabetical)	1
Subtotal	**2**

Model answer:

```
SELECT StudentName, Department
FROM STUDENT
WHERE Department = 'ICT'
ORDER BY StudentName ASC;
```

b) (2 marks)

Answer	Marks
SELECT CourseID, AVG(Marks) FROM ENROLLMENT GROUP BY CourseID	1
HAVING AVG(Marks) > 60 (HAVING, not WHERE)	1
Subtotal	**2**

Model answer:

```
SELECT CourseID, AVG(Marks) AS AverageMarks
```

```
FROM ENROLLMENT

GROUP BY CourseID

HAVING AVG(Marks) > 60;
```

c) (2 marks)

Answer	Marks
Selects the correct columns (StudentName, CourseID, Marks) from both tables	1
Correct join condition ON STUDENT.StudentID = ENROLLMENT.StudentID (or WHERE equivalent)	1
Subtotal	**2**

Model answer:

```
SELECT S.StudentName, E.CourseID, E.Marks

FROM STUDENT S

INNER JOIN ENROLLMENT E ON S.StudentID = E.StudentID;
```

Question 9 Total 6